EXPERIMENT 4

Protocols Learnings in detail using Wireshark filter and flow Graph

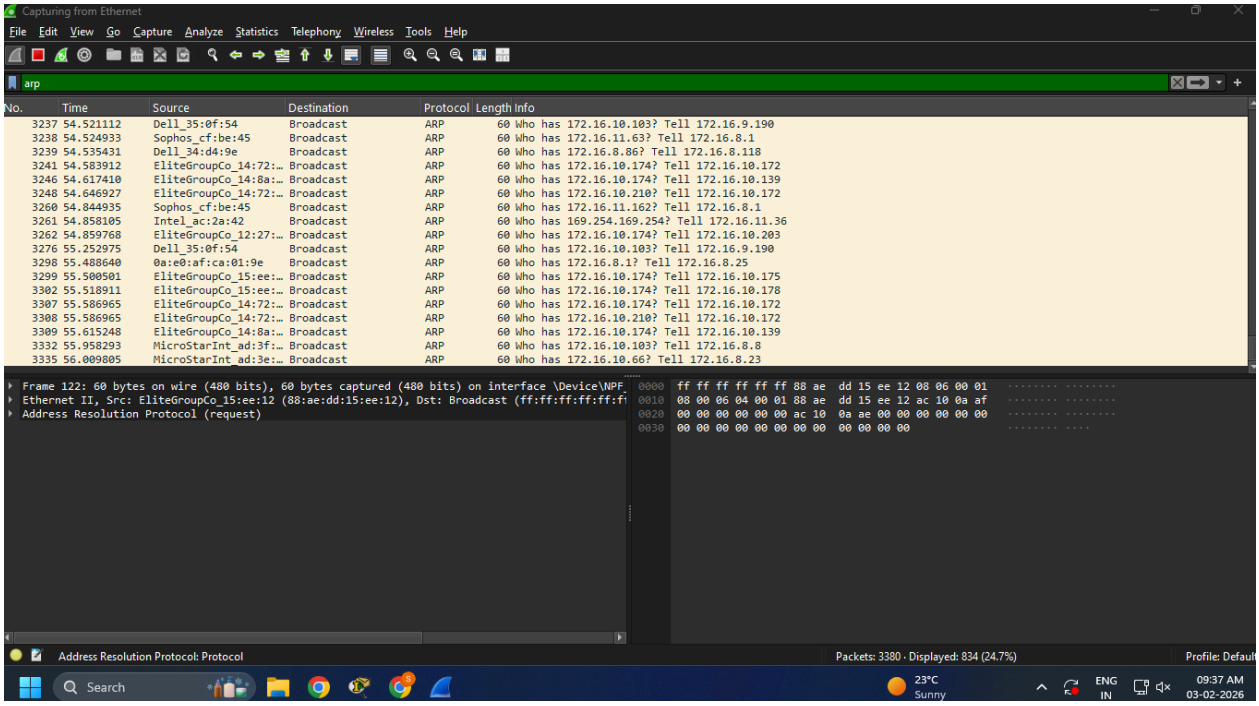
AIM:

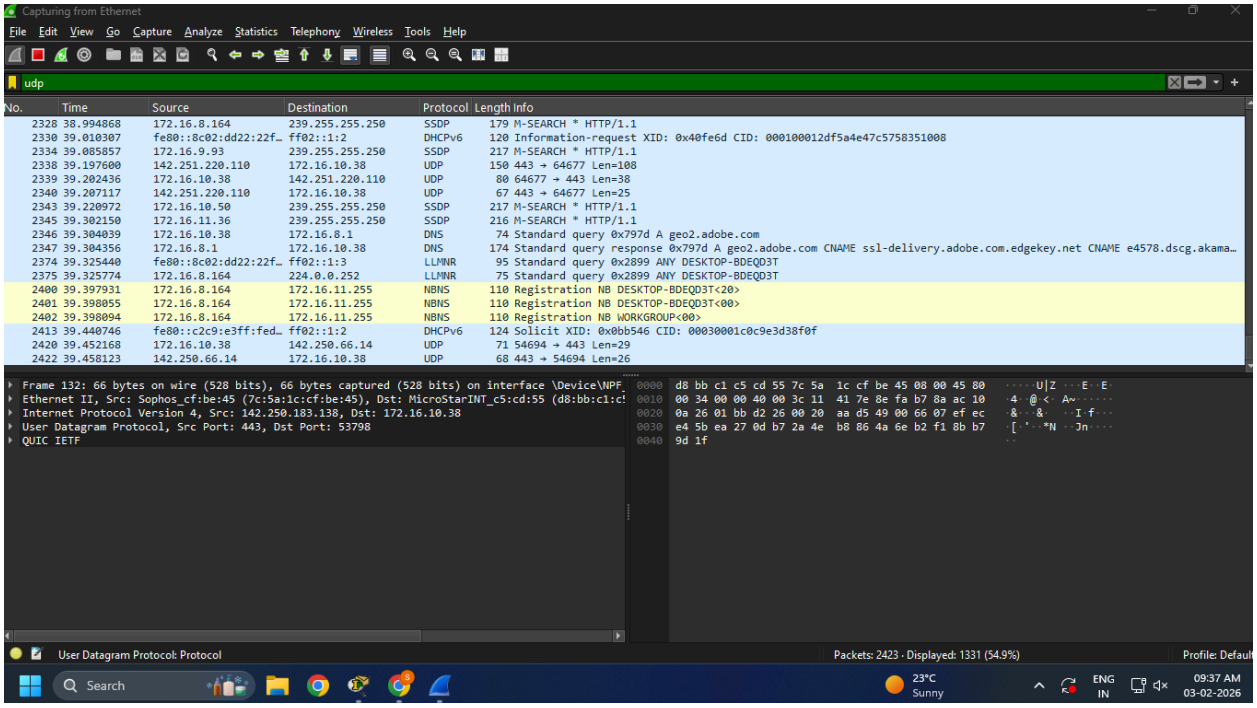
To study various network protocols and analyze their working using Wireshark by applying display filters and observing flow graphs.

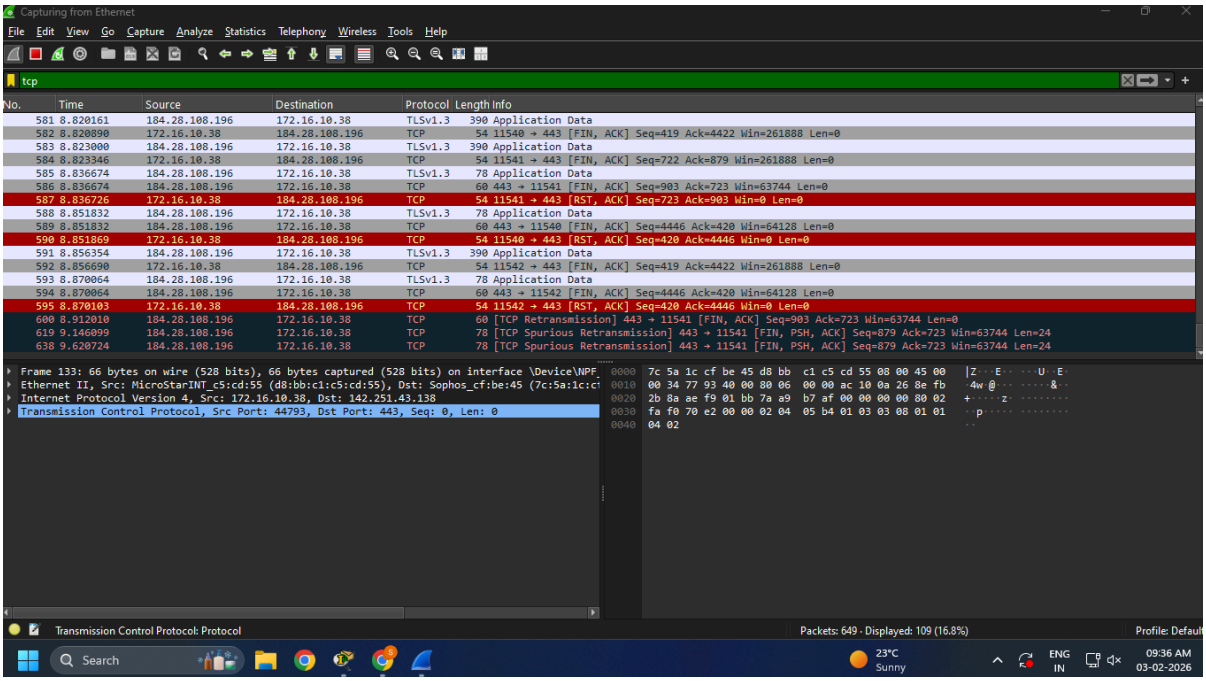
PROCEDURE:

1. Install and open Wireshark.
2. Select the active network interface (Wi-Fi / Ethernet).
3. Click Start Capture.
4. Perform network activities:
    - Open a website
    - Ping a server
    - Use any online application
5. Stop the capture after a few minutes.
6. Apply protocol display filters in the filter bar.
7. Select packets → Go to
   Statistics → Flow Graph
8. Observe packet flow, sequence, and communication between hosts.

OUTPUT:


Capturing from Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
arp
No. Time Source Destination Protocol Length Info
3237 54.521112 Dell_35:0f:54 Broadcast ARP 60 Who has 172.16.10.103? Tell 172.16.9.190
3238 54.524933 Sophos_cf:be:45 Broadcast ARP 60 Who has 172.16.11.63? Tell 172.16.8.1
3239 54.535431 Dell_34:d4:9e Broadcast ARP 60 Who has 172.16.8.86? Tell 172.16.8.118
3241 54.583912 EliteGroupCo_14:72:… Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.172
3246 54.617410 EliteGroupCo_14:8a:… Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.139
3248 54.646927 EliteGroupCo_14:72:… Broadcast ARP 60 Who has 172.16.10.210? Tell 172.16.10.172
3260 54.844935 Sophos_cf:be:45 Broadcast ARP 60 Who has 172.16.11.162? Tell 172.16.8.1
3261 54.858105 Intel_ac:2a:42 Broadcast ARP 60 Who has 169.254.169.254? Tell 172.16.11.36
3262 54.859768 EliteGroupCo_12:27:… Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.203
3276 55.252975 Dell_35:0f:54 Broadcast ARP 60 Who has 172.16.10.103? Tell 172.16.9.190
3298 55.488640 0a:e0:af:ca:01:9e Broadcast ARP 60 Who has 172.16.8.1? Tell 172.16.8.25
3299 55.500501 EliteGroupCo_15:ee:… Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.175
3302 55.518911 EliteGroupCo_15:ee:… Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.178
3307 55.586965 EliteGroupCo_14:72:… Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.172
3308 55.586965 EliteGroupCo_14:72:… Broadcast ARP 60 Who has 172.16.10.210? Tell 172.16.10.172
3309 55.615248 EliteGroupCo_14:8a:… Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.139
3332 55.958293 MicroStarInt_ad:3f:… Broadcast ARP 60 Who has 172.16.10.103? Tell 172.16.8.8
3335 56.009805 MicroStarInt_ad:3e:… Broadcast ARP 60 Who has 172.16.10.66? Tell 172.16.8.23
Frame 122: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_
Ethernet II, Src: EliteGroupCo_15:ee:12 (88:ae:dd:15:ee:12), Dst: Broadcast (ff:ff:ff:ff:ff:f
Address Resolution Protocol (request)
0000 ff ff ff ff ff ff 88 ae dd 15 ee 12 08 06 00 01
0010 08 00 06 04 00 01 88 ae dd 15 ee 12 ac 10 0a af
0020 00 00 00 00 00 00 ac 10 0a ae 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00
Address Resolution Protocol: Protocol
Packets: 3380 · Displayed: 834 (24.7%)
Profile: Default
Search
23°C Sunny
ENG IN
09:37 AM 03-02-2026


Capturing from Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
2328 38.994868 172.16.8.164 239.255.255.250 SSDP 179 M-SEARCH * HTTP/1.1
2330 39.010307 fe80::8c02:dd22:22f… ff02::1:2 DHCPv6 120 Information-request XID: 0x40fe6d CID: 000100012df5a4e47c5758351008
2334 39.085857 172.16.9.93 239.255.255.250 SSDP 217 M-SEARCH * HTTP/1.1
2338 39.197600 142.251.220.110 172.16.10.38 UDP 150 443 → 64677 Len=108
2339 39.202436 172.16.10.38 142.251.220.110 UDP 80 64677 → 443 Len=38
2340 39.207117 142.251.220.110 172.16.10.38 UDP 67 443 → 64677 Len=25
2343 39.220972 172.16.10.50 239.255.255.250 SSDP 217 M-SEARCH * HTTP/1.1
2345 39.302150 172.16.11.36 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
2346 39.304039 172.16.10.38 172.16.8.1 DNS 74 Standard query 0x797d A geo2.adobe.com
2347 39.304356 172.16.8.1 172.16.10.38 DNS 174 Standard query response 0x797d A geo2.adobe.com CNAME ssl-delivery.adobe.com.edgekey.net CNAME e4578.dscg.akama…
2374 39.325440 fe80::8c02:dd22:22f… ff02::1:3 LLMNR 95 Standard query 0x2899 ANY DESKTOP-BDEQD3T
2375 39.325774 172.16.8.164 224.0.0.252 LLMNR 75 Standard query 0x2899 ANY DESKTOP-BDEQD3T
2400 39.397931 172.16.8.164 172.16.11.255 NBNS 110 Registration NB DESKTOP-BDEQD3T<20>
2401 39.398055 172.16.8.164 172.16.11.255 NBNS 110 Registration NB DESKTOP-BDEQD3T<00>
2402 39.398094 172.16.8.164 172.16.11.255 NBNS 110 Registration NB WORKGROUP<00>
2413 39.440746 fe80::c2c9:e3ff:fed… ff02::1:2 DHCPv6 124 Solicit XID: 0x0bb546 CID: 00030001c0c9e3d38f0f
2420 39.452168 172.16.10.38 142.250.66.14 UDP 71 54694 → 443 Len=29
2422 39.458123 142.250.66.14 172.16.10.38 UDP 68 443 → 54694 Len=26
Frame 132: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_
Ethernet II, Src: Sophos_cf:be:45 (7c:5a:1c:cf:be:45), Dst: MicroStarINT_c5:cd:55 (d8:bb:c1:c5
Internet Protocol Version 4, Src: 142.250.183.138, Dst: 172.16.10.38
User Datagram Protocol, Src Port: 443, Dst Port: 53798
QUIC IETF
0000 d8 bb c1 c5 cd 55 7c 5a 1c cf be 45 08 00 45 80
0010 00 34 00 00 40 00 3c 11 41 7e 8e fa b7 8a ac 10
0020 0a 26 01 bb d2 26 00 20 aa d5 49 00 66 07 ef ec
0030 e4 5b ea 27 0d b7 2a 4e b8 86 4a 6e b2 f1 8b b7
0040 9d 1f
User Datagram Protocol: Protocol
Packets: 2423 · Displayed: 1331 (54.9%)
Profile: Default
Search
23°C Sunny
ENG IN
09:37 AM 03-02-2026



RESULT:

Thus, various network protocols such as TCP, UDP, ARP, ICMP, and DNS were successfully captured and analyzed using Wireshark.